DT 111 · TOPIC 1

Introduction to Data Science and Artificial Intelligence

Foundations, relationships, workflow, history, applications and limitations

Mkwawa University College of Education

BSc Data Science and Artificial Intelligence

Learning outcomes

- Explain Data Science and Artificial Intelligence in clear terms.
- Distinguish Data Science, AI and Machine Learning.
- Describe a simple Data Science workflow from problem to decision.
- Recognise major milestones and real-world applications.
- Discuss benefits, limitations and responsible use.

Data Science combines several disciplines

Data Science is not simply “using Python” or “making charts”.

Data

Recorded observations, measurements, text, images or signals.

Statistics

Tools for summarising uncertainty, variation and relationships.

Computing

Storage, programming, algorithms and scalable processing.

Domain knowledge

Context needed to ask meaningful questions and interpret results.

From raw data to a useful decision

Evidence becomes valuable when it is connected to a meaningful question.

1

Data

45, 62, 81, 39, 73

2

Information

Mean score = 60

3

Insight

Some students may need support

4

Question

Is attendance related?

5

Action

Investigate and support

AI and ordinary automation

Automation

- Follows predefined instructions
- Excellent for routine, explicit processes
- Example: calculate the average of five marks

Artificial Intelligence

- Uses methods associated with reasoning, learning, perception or language
- May adapt from data or search among alternatives
- Example: recognise crop disease from an image

Data Science, AI and Machine Learning

The fields overlap, but they are not identical.

Data Science

Uses data, statistics, computing and domain knowledge to understand problems and support decisions.

Artificial Intelligence

Broad field concerned with systems that perform tasks associated with intelligent behaviour.

Machine Learning

A major AI approach in which models learn patterns from examples or experience.

Four types of analytical questions

DESCRIPTIVE

What happened?

Average score, attendance rate, number of disease cases.

DIAGNOSTIC

What may explain it?

Compare groups and investigate relationships.

PREDICTIVE

What may happen next?

Estimate an unknown or future outcome.

PRESCRIPTIVE

What action should be considered?

Combine evidence, constraints and human judgement.

A simple Data Science workflow

1. **Problem**
   Define the question
2. **Data**
   Collect relevant evidence
3. **Prepare**
   Clean and transform
4. **Explore**
   Summarise and visualise
5. **Model / analyse**
   Find patterns or predict
6. **Communicate**
   Interpret and support action

A model is a simplified representation

A model is useful when it is appropriate for the question and sufficiently reliable.

Equation

A mathematical relationship between variables.

Rules

IF-THEN logic that represents explicit knowledge.

Tree

A sequence of questions that leads to an output.

Neural network

A learned function built from connected weighted units.

History and evolution

Modern Data Science and AI emerged from decades of statistics, computing, algorithms and increasing access to data.

A short evolution timeline

1

1950s

Early AI and symbolic reasoning

2

1980s

Expert systems and knowledge bases

3

1990s-2000s

Data mining and statistical learning

4

2010s

Deep learning grows with data and GPUs

5

2020s

Generative and multimodal AI

Why AI accelerated

More data

Digital services, sensors, images, text and online activity generate large datasets.

More computing power

GPUs, cloud computing and specialised hardware make larger models practical.

Improved algorithms

Advances in optimisation, neural networks and software frameworks improve capability.

Applications across sectors

Agriculture

Crop monitoring, disease recognition, yield forecasting.

Health

Medical imaging, triage support, disease-risk analysis.

Finance

Fraud detection, credit-risk support, forecasting.

Education

Learning analytics, personalised support, automated assistance.

Transport

Route planning, traffic analysis, driver-assistance systems.

Public services

Planning, resource allocation and citizen-support systems.

Example: crop-disease decision support

Errors can enter at data capture, prediction or use of the output.

1

Image

Farmer captures leaf photo

2

Model

Image features are analysed

3

Prediction

Likely disease / stage

4

Context

Confidence and agronomic knowledge

5

Decision

Advice or referral

Benefits and limitations

Potential benefits

- Speed and automation
- Pattern discovery in large datasets
- Consistent support for repeated tasks
- New tools for science and services

Important limitations

- Data can be incomplete or biased
- Models can make errors
- Correlation does not guarantee cause
- Privacy and fairness require attention

Not every problem needs AI

Choose technology from the problem, not from popularity.

Average five marks

A simple arithmetic algorithm is enough.

Store student records

A database solves the main problem.

Recognise disease from images

Machine Learning may be appropriate.

Topic 1 key ideas

Data Science connects data, methods, domain knowledge and decisions.

AI is broader than Machine Learning; Machine Learning is one major AI approach.

A good workflow begins with the problem and ends with interpretation and action.

Models are simplified representations, not perfect copies of reality.

Useful AI requires evidence, evaluation and responsible human judgement.